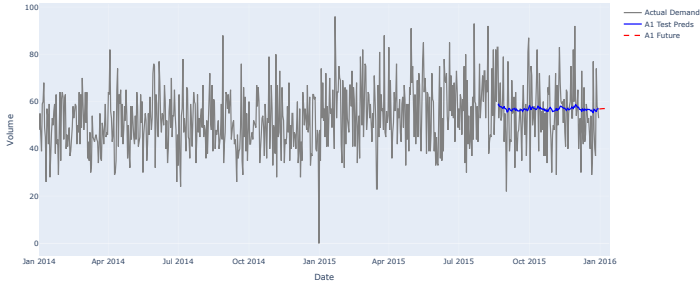


A (LSTM) : [64h/L1 | 128h/L2] [LB:2 | LB:3]
B (SRMX) : [(1,1,0)| (2,1,2)] [None | 6mo-S]
C (XGB) : [Depth 6 | Depth 8] [Depth 4 | Depth 6]
E (RECO) : [64e/128h | 128e/256h] (No Monthly)

* *h=Hidden, L=Layers, LB=Lookback, S=Season, e=Embed*

Run A1 - Demand Forecasting Inference



```
# Run Overview: A1
##### Tunable hyperparameters

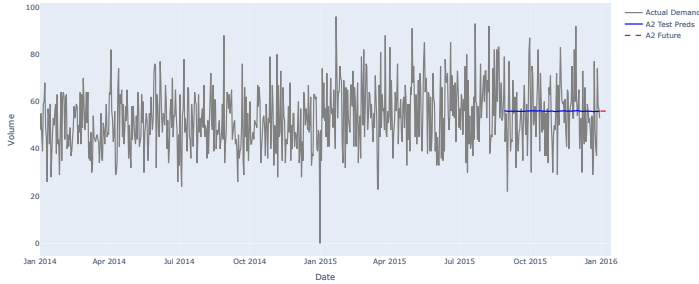
| hidden_size | num_layers | learning_rate | batch_size | epochs | lookback_window | future_steps | model_type | freq | loss_type |
| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 64 | 1 | 0.001 | 16 | 30 | 14 | 14 | forecasting_lstm | D | MSE |

##### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 10.6113 | 13.5388 | 0.0762 |
```

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Run A2 - Demand Forecasting Inference



```
# Run Overview: A2
##### Tunable hyperparameters

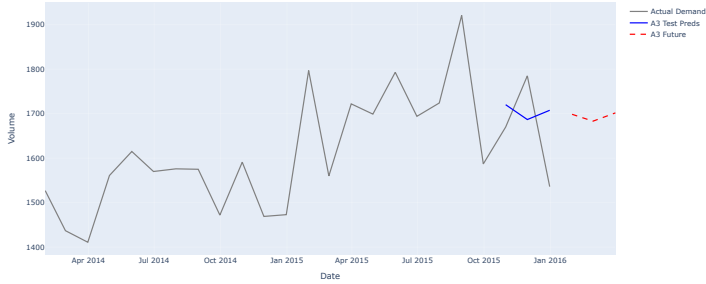
| hidden_size | num_layers | learning_rate | batch_size | epochs | lookback_window | future_steps | model_type | freq | loss_type |
| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 128 | 2 | 0.001 | 16 | 50 | 21 | 14 | forecasting_lstm | D | MSE |

##### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 10.6194 | 13.556 | 0.2154 |
```

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Run A3 - Demand Forecasting Inference



```
# Run Overview: A3
##### Tunable hyperparameters

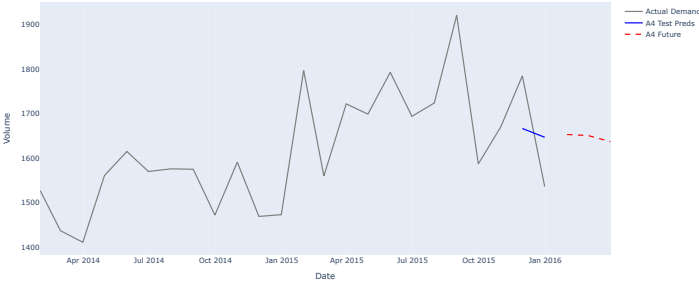
| hidden_size | num_layers | learning_rate | batch_size | epochs | lookback_window | future_steps | model_type | freq | loss_type |
| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 64 | 1 | 0.001 | 16 | 30 | 2 | 3 | forecasting_lstm | ME | Huber |

##### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 106.6597 | 117.7701 | 0.0027 |
```

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Run A4 - Demand Forecasting Inference



```
# Run Overview: A4
##### Tunable hyperparameters

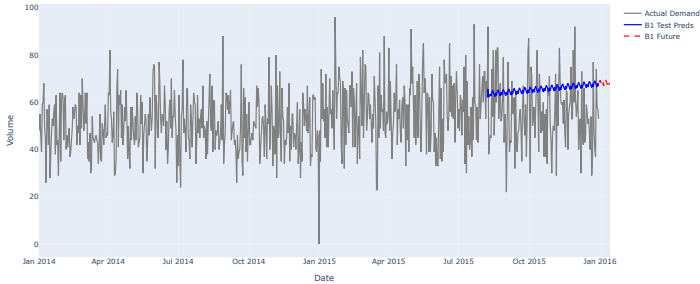
| hidden_size | num_layers | learning_rate | batch_size | epochs | lookback_window | future_steps | model_type | freq | loss_type |
| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 128 | 2 | 0.001 | 16 | 50 | 3 | 3 | forecasting_lstm | ME | Huber |

##### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 114.6422 | 114.7015 | 0.0052 |
```

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Run B1 - Demand Forecasting Inference



```
# Run Overview: B1
##### Tunable hyperparameters

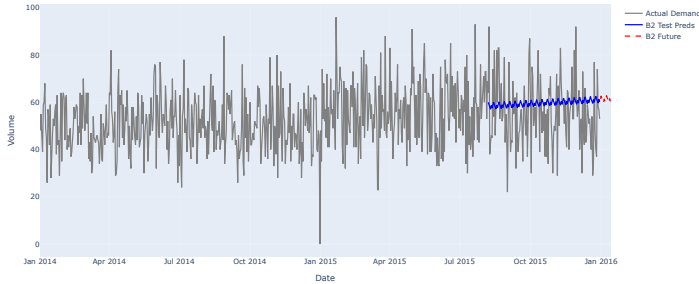
| order | seasonal_order | future_steps | model_type | freq |
| --- | --- | --- | --- | --- |
| (1, 1, 0) | (0, 1, 1, 7) | 14 | forecasting_sarimax | D |

##### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 14.6661 | 17.4371 | 0.0042 |
```

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Run B2 - Demand Forecasting Inference



```
# Run Overview: B2
##### Tunable hyperparameters

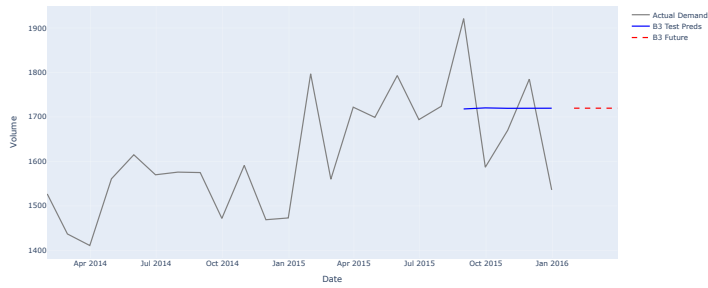
| order | seasonal_order | future_steps | model_type | freq |
| --- | --- | --- | --- | --- |
| (2, 1, 2) | (1, 1, 1, 7) | 14 | forecasting_sarimax | D |

##### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 11.876 | 14.7411 | 0.037 |
```

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Run B3 - Demand Forecasting Inference



Run Overview: B3

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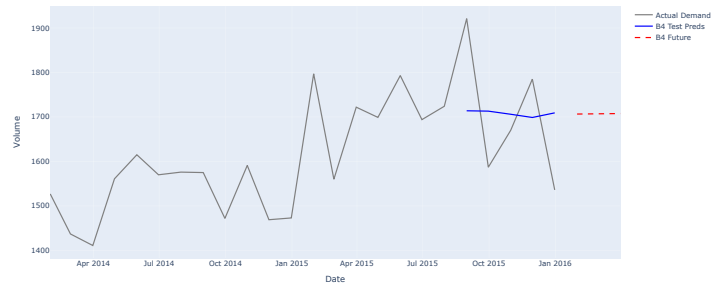
```
#### Tunable hyperparameters

| order | seasonal_order | future_steps | model_type | freq |
| --- | --- | --- | --- | --- |
| (1, 1, 1) | (0, 0, 0, 0) | 3 | forecasting_sarimax | ME |

#### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 126.9315 | 141.0222 | 0.0014 |
```

Run B4 - Demand Forecasting Inference



Run Overview: B4

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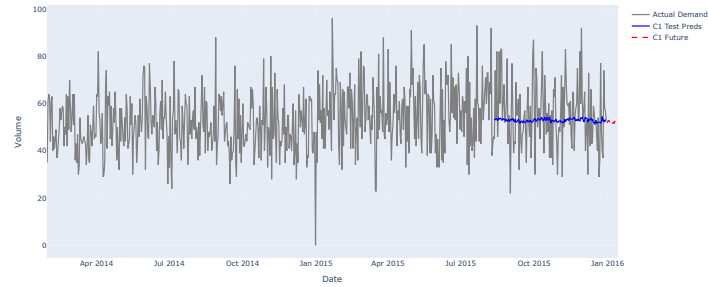
```
#### Tunable hyperparameters

| order | seasonal_order | future_steps | model_type | freq |
| --- | --- | --- | --- | --- |
| (1, 1, 1) | (1, 0, 0, 6) | 3 | forecasting_sarimax | ME |

#### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 125.7315 | 139.6222 | 0.0018 |
```

Run C1 - Demand Forecasting Inference



Run Overview: C1

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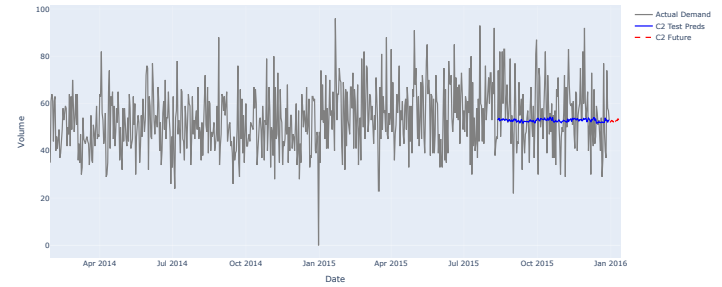
```
#### Tunable hyperparameters

| n_estimators | learning_rate | max_depth | subsample | colsample_bytree | early_stopping_rounds | future_steps | freq | model_type |
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 1000 | 0.01 | 6 | 0.8 | 0.8 | 100 | 14 | D | forecasting_xgboost |

#### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 10.6865 | 13.8227 | 0.0036 |
```

Run C2 - Demand Forecasting Inference



Run Overview: C2

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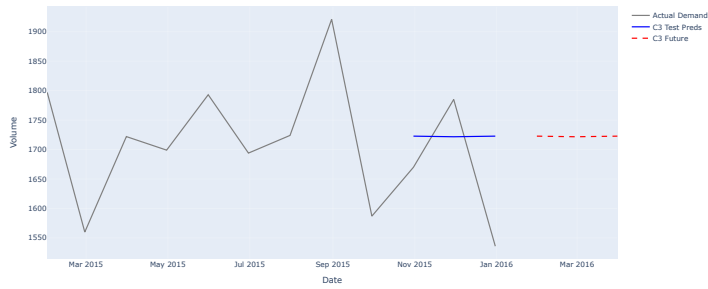
```
#### Tunable hyperparameters

| n_estimators | learning_rate | max_depth | subsample | colsample_bytree | early_stopping_rounds | future_steps | freq | model_type |
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 1000 | 0.01 | 8 | 0.8 | 0.8 | 100 | 14 | D | forecasting_xgboost |

#### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 10.7057 | 13.8937 | 0.0095 |
```

Run C3 - Demand Forecasting Inference



Run Overview: C3

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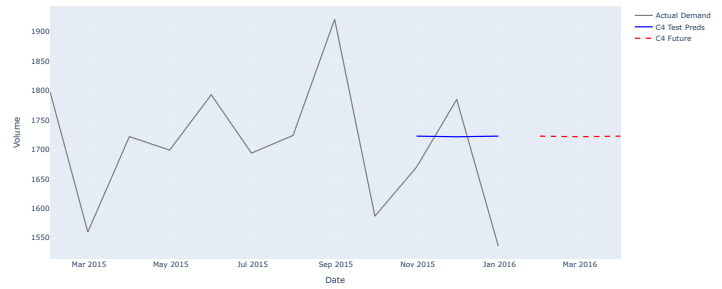
```
#### Tunable hyperparameters

| n_estimators | learning_rate | max_depth | subsample | colsample_bytree | early_stopping_rounds | future_steps | freq | model_type |
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 1000 | 0.01 | 4 | 0.8 | 0.8 | 100 | 3 | ME | forecasting_xgboost |

#### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 100.9223 | 117.8468 | 0.0038 |
```

Run C4 - Demand Forecasting Inference



Run Overview: C4

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```
#### Tunable hyperparameters

| n_estimators | learning_rate | max_depth | subsample | colsample_bytree | early_stopping_rounds | future_steps | freq | model_type |
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 1000 | 0.01 | 6 | 0.8 | 0.8 | 100 | 3 | ME | forecasting_xgboost |

#### Evaluation Metrics

| mae | rmse | elapsed_time_min |
| --- | --- | --- |
| 100.9223 | 117.8468 | 0.004 |
```